

TengeFlow

Personal expense tracking
for everyday life in Kazakhstan.

Know where your money goes.

Pinigin Artyom / BDA-2602

Information and Communication Technologies
Midterm Project, Stage 1 / Topic 35



The problem and the person

Small payments. An unclear monthly picture.

Card and cash spending are easy to forget. A transaction list alone does not show how each purchase affects a personal plan.

The need: quick entry, understandable categories and an immediate remaining budget.

Design basis: the project brief and CFPB spending-tracker guidance [1].

Ayan, 20

Illustrative persona / university student

Goal

Manage scholarship and family support without learning accounting.

Context

Smartphone daily, laptop for study.
Medium-high digital literacy.

Five screens, one clear purpose

01 Overview

See what remains this month.

02 Add Expense

Primary task: record a purchase.

03 Transactions

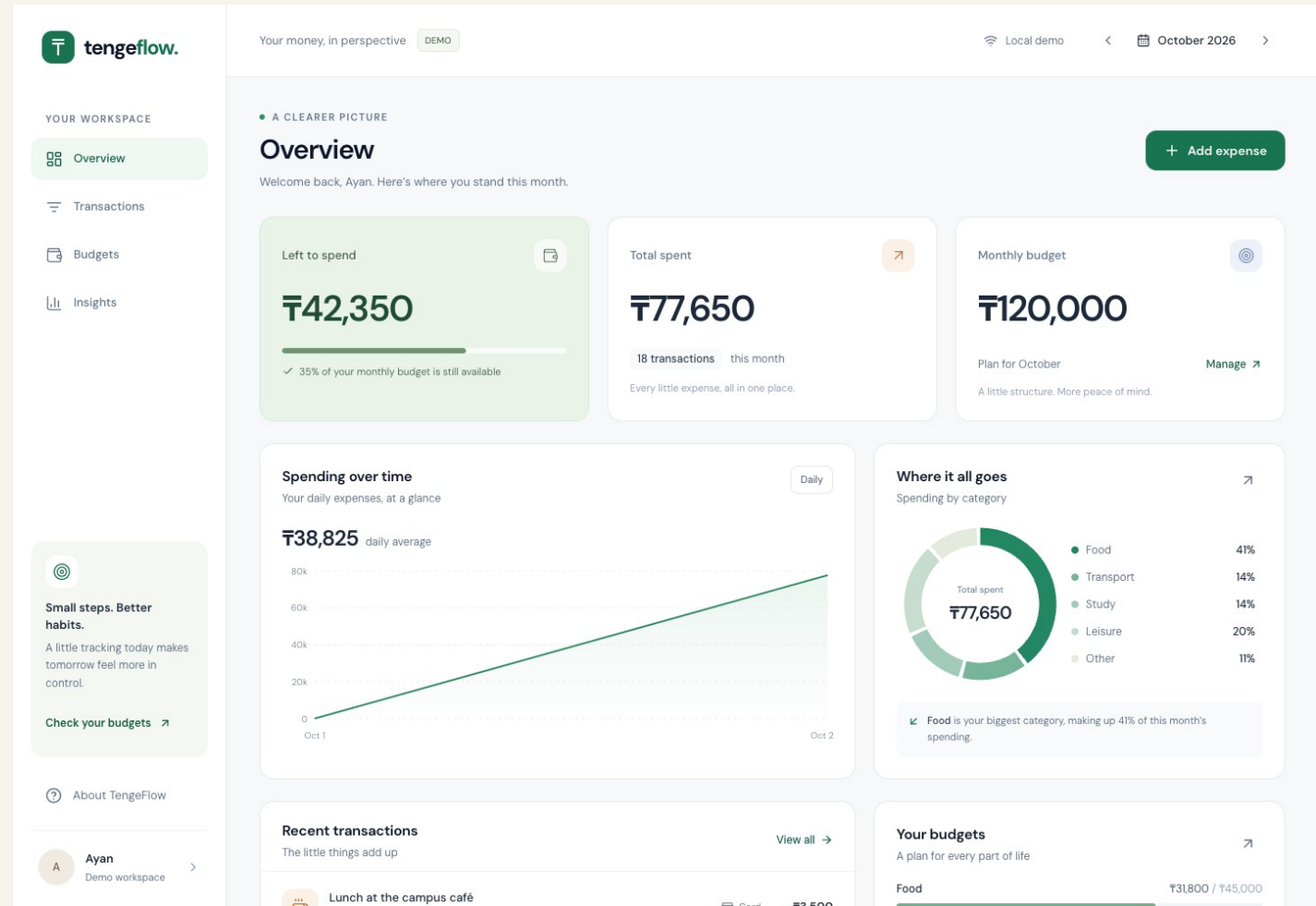
Find, edit or delete a record.

04 Budgets

Set total and category limits.

05 Insights

Review categories and trends.



Actual demo interface. Seed data is fictional.

From registration to a personal budget

 [← Back to home](#)

START WITH A LITTLE CLARITY

A fresh start for your money.

Create your personal TengeFlow workspace.

Your name

Email address

Password

Confirm password

Create account →

Already have an account? [Sign in](#)

or take a look first

 **tengeflow.** [\[→ Sign out\]](#)

A LITTLE PLANNING. A FRESH START.

**Make room for
*your kind of life.***

A few small choices now.
A clearer picture of your month ahead.



Your plans can change. Your budget can, too.

✓ About you

2 Your month

3 The little things

STEP 2 OF 3

**Give your month
*a little direction.***

Choose an amount that feels right for your month. You can always change it later.

Your monthly budget

KZT

This is your spending plan, not an account balance.

₹60,000

₹120,000

₹200,000

Pick a starting point above, or enter your own amount. Next, you'll decide how to split it.

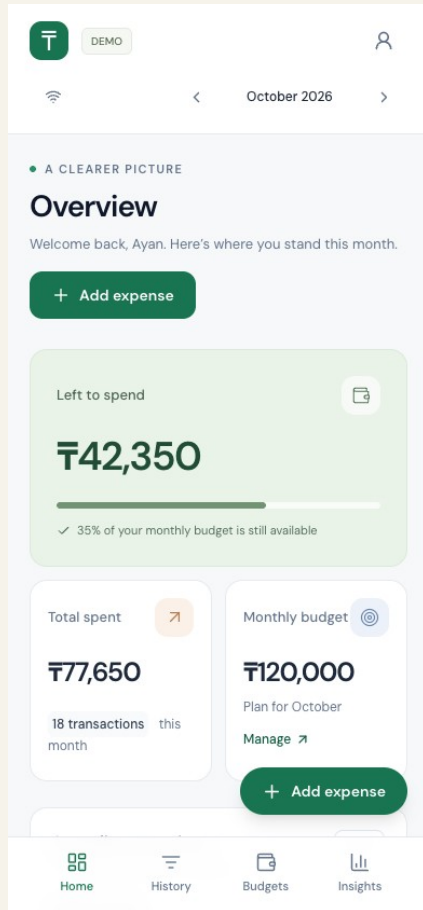
[← Back](#) [Continue →](#)

You can adjust your budget and limits later.

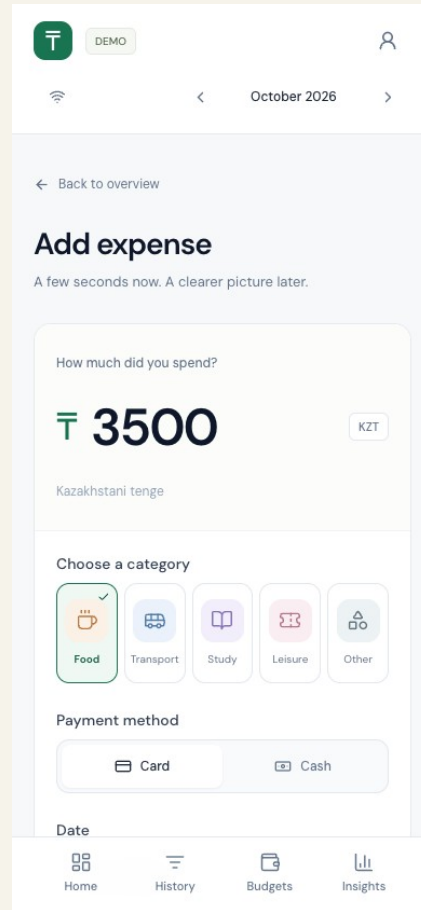
Register → confirm email → set up your workspace

Setup collects a name, monthly budget and category limits. A new account starts with no expenses.

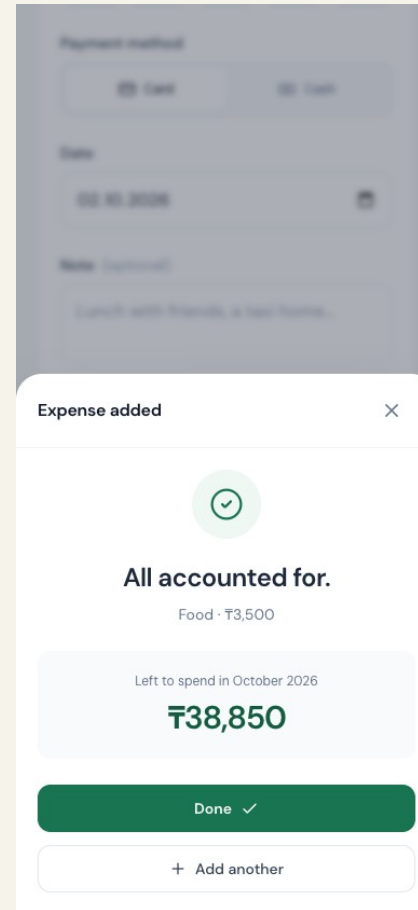
A purchase becomes budget feedback



1 Start



2 Record



3 Confirm

Campus lunch

3,500 KZT
Food · Card · Today

42,350

KZT left before

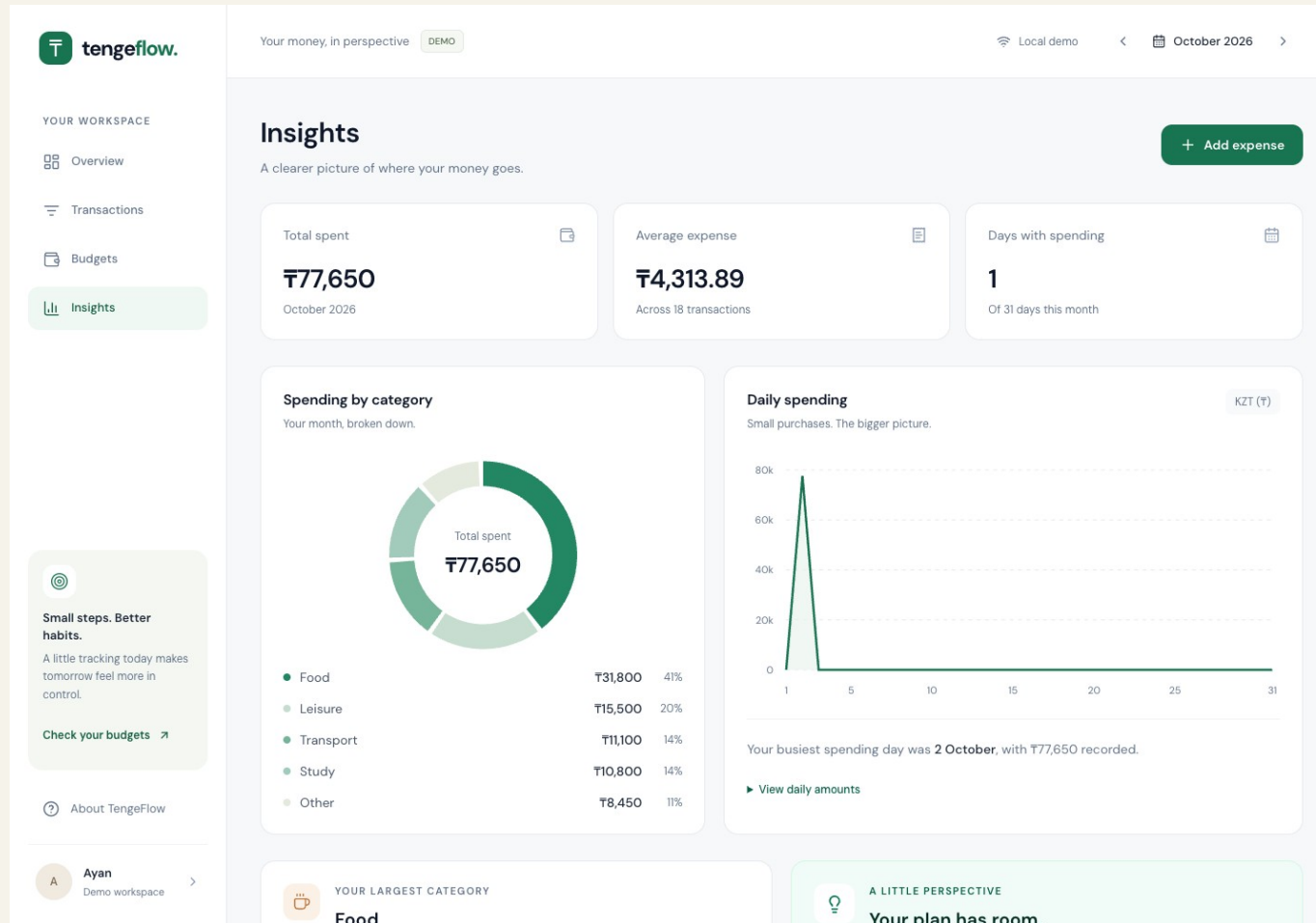
38,850

KZT left after saving

Illustrative demo: 120,000 KZT budget – 81,150 KZT spent = 38,850 KZT remaining.

TengeFlow / ICT Midterm

Review spending and adjust the plan



Fictional demo data. The charts describe recorded spending, not forecasts.

Find a pattern

Category totals and a spending trend explain where money goes.

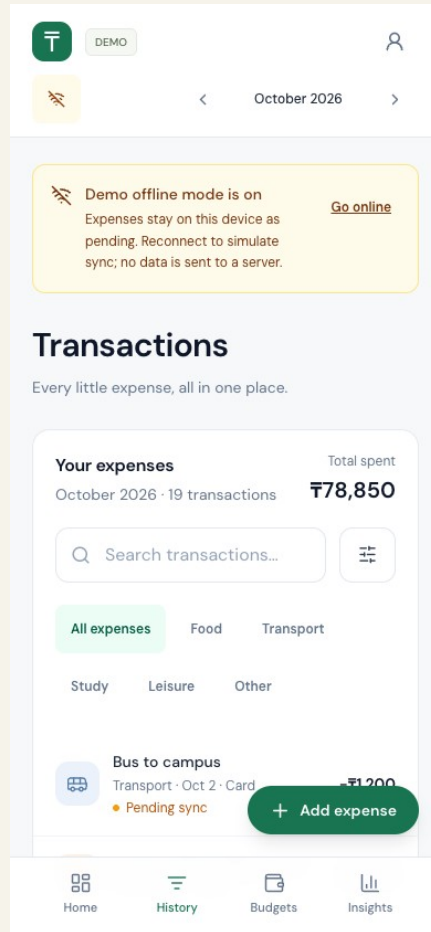
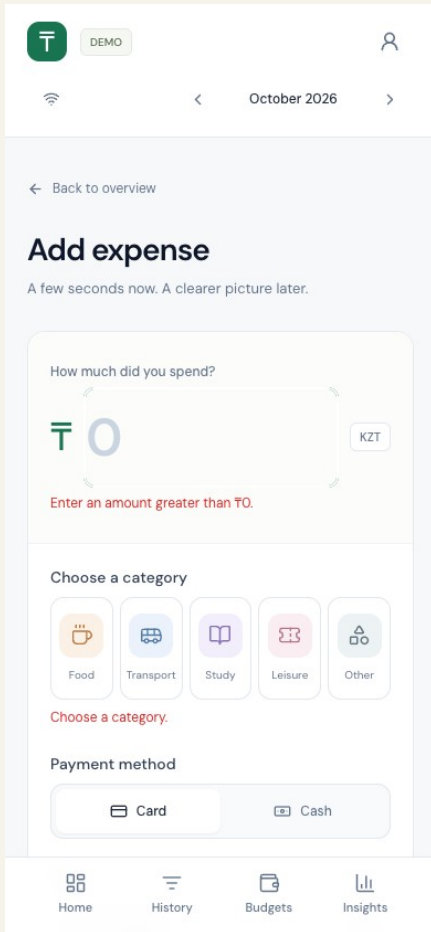
Check the details

Search and filter history, then correct an expense if needed.

Adjust the plan

Change limits and see warnings when a budget is exceeded.

Mobile interaction and recovery



Errors explain the next step.

Inline messages identify invalid fields. Drafts remain available so the user can correct the form and retry.

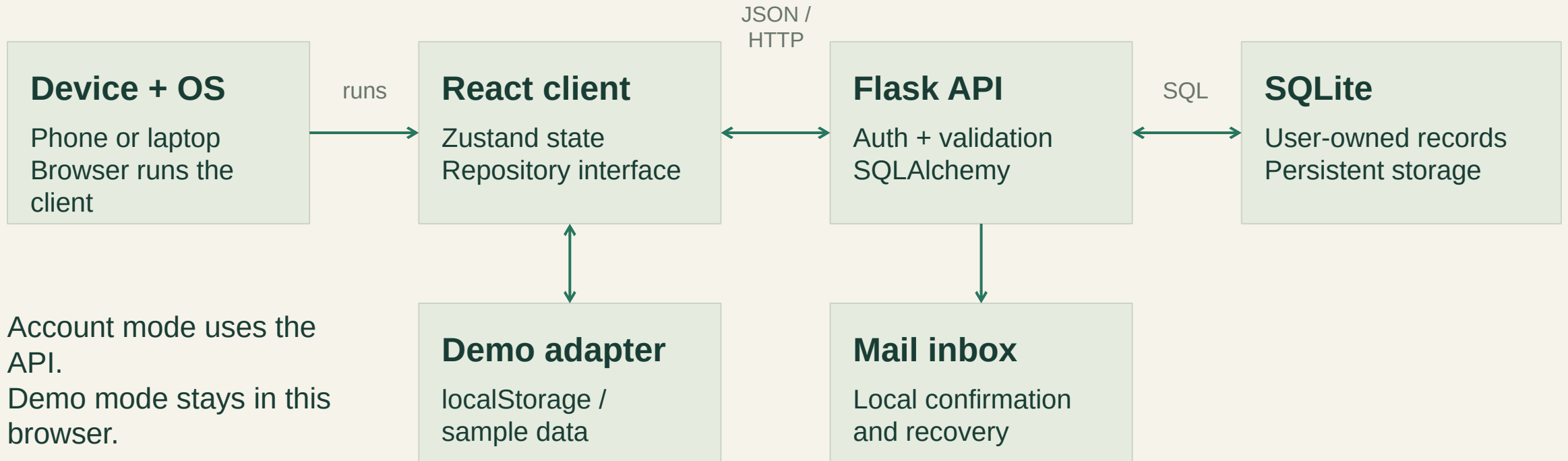
Designed for a phone.

Readable amounts, touch-sized controls and bottom navigation support everyday use. Dialogs manage focus and respect reduced motion.

Demo pending status is simulated. Account saves need a connection.

System architecture

Implemented locally



Internet deployment

Wi-Fi / 4G / 5G → HTTPS → Vercel frontend
Account services: Flask API → database + email
Frontend demo is live. The account backend runs locally.

One complete data transaction

1 Enter and validate

The form checks amount, category and date. Invalid input stays on the device.

2 Send the request

The repository sends JSON and an access token to POST /api/transactions.

3 Check permission

Flask verifies the session, validates the fields and checks category ownership.

4 Commit the expense

SQLAlchemy stores an expense owned by the authenticated user.

5 Return the saved record

The API responds with the saved data and a server-generated ID.

6 Update the interface

Zustand updates state. Totals recalculate and the success dialog appears.

A failed account request keeps the draft and shows an error. Success is shown after the server confirms the save.

Mobile and cloud trade-offs

A responsive web app uses one interface on phones and laptops.

	Connected account architecture	Fully local tracker
Access	Designed for the same account on multiple devices.	Data stays in one browser or on one device.
Storage	Server owns saved records. Backups require an operating plan.	No remote service dependency. Local data can be lost.
Connectivity	Account requests need a network. Use HTTPS when hosted.	Stored records work locally, if the app itself can load.
Trade-off	Hosting cost, outages and privacy need ongoing attention.	Limited device continuity and recovery options.

Wi-Fi or mobile data carries API requests. Local form validation and demo data stay in the browser.

Limitations and next development

Current boundary	Next practical step
Manual entry, KZT and five fixed categories.	Add recurring expenses and CSV export.
Account writes stop without a connection. The demo only simulates synchronization.	Design a real offline queue with duplicate prevention and conflict handling.
The frontend demo is public. Account services and email run locally.	Host the Flask API, configure SMTP email, and test database backup and recovery.
The quick-entry goal has not been measured. No student usability study has been completed.	Test the flow with students and measure completion time, errors and recovery.

The scope is expense tracking. Bank linking and payment execution are outside this prototype.

TengeFlow

A recorded purchase.

A clearer personal budget.

The core flow is ready to demonstrate:

Overview → Add Expense → Confirmation → Updated budget

Thank you. Questions?

Transaction history

tengeflow.

YOUR WORKSPACE

Overview

Transactions

Budgets

Insights

Small steps. Better habits.

A little tracking today makes tomorrow feel more in control.

Check your budgets

About TengeFlow

Ayan

Demo workspace

Your money, in perspective

DEMO

Local demo

October 2026

Transactions

Every little expense, all in one place.

+ Add expense

Your expenses

October 2026 · 18 transactions

Total spent
₹77,650

Search transactions...

Filters

All expenses

Food

Transport

Study

Leisure

Other

Transaction	Category	Date	Payment	Amount
 Lunch at the campus café	Food	Oct 2	 Card	-₹3,500
 Bus / taxi	Transport	Oct 2	 Card	-₹1,200
 Printing	Study	Oct 2	 Cash	-₹2,450
 Cinema with friends	Leisure	Oct 2	 Card	-₹5,500
 Groceries	Food	Oct 2	 Card	-₹8,200
 Mobile plan	Other	Oct 2	 Card	-₹4,990
 Coffee and a sandwich	Food	Oct 2	 Card	-₹2,800

Search and filter

Find a record by note, category, payment method or period.

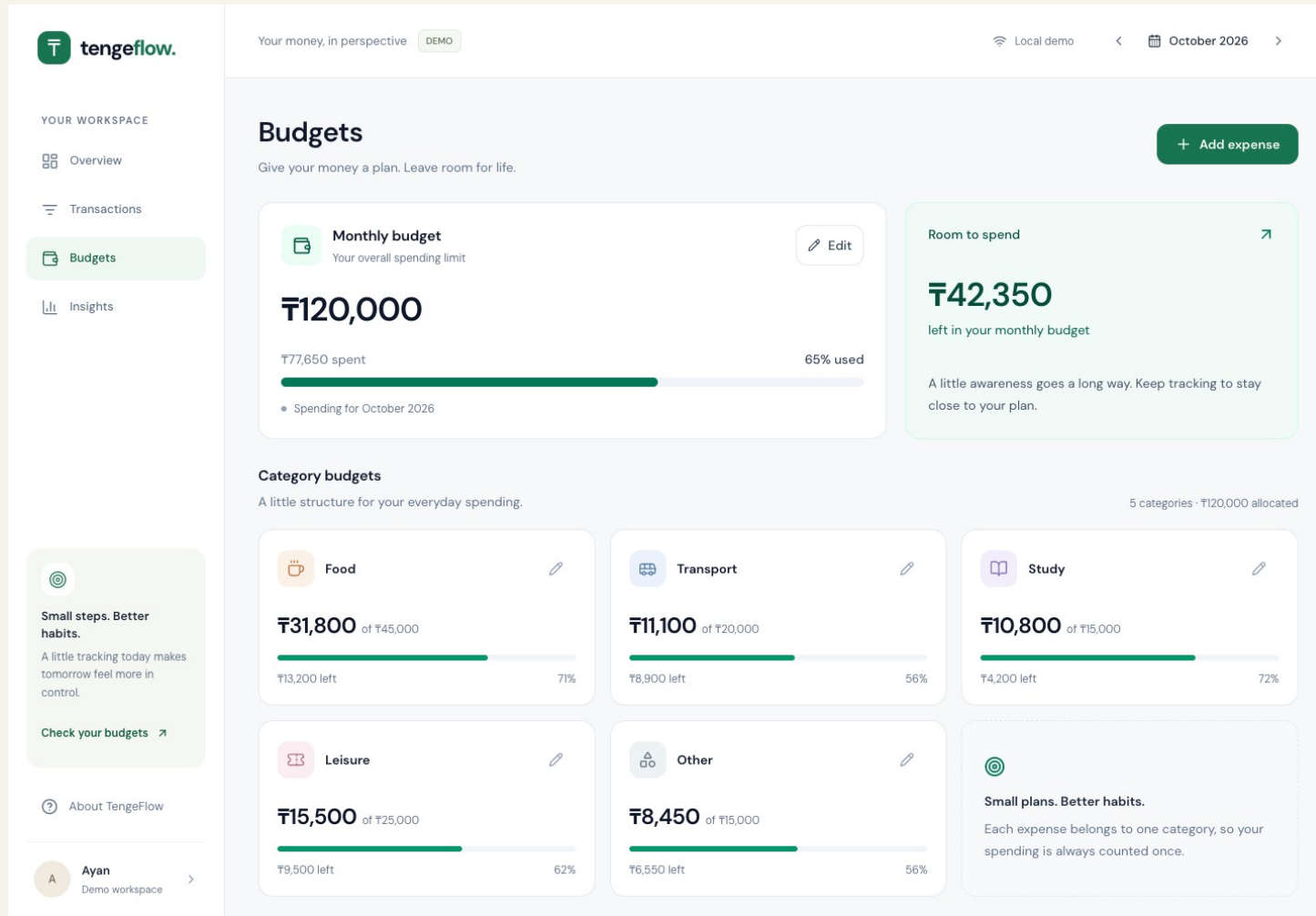
Correct mistakes

Edit the amount and details. Confirm before deleting an expense.

Keep totals consistent

Changes feed back into the overview, budgets and insights.

Category budgets



Category limits

Food, Transport, Study, Leisure and Other have separate limits.

No double counting

Each transaction belongs to one category and affects one category total.

Visible budget status

Compare spending with the limit and show an overspending warning.

Private accounts and reliable records

Authentication

Passwords use scrypt hashing. Short-lived access JWTs stay in memory.

Session continuity

A rotating refresh token uses an HttpOnly cookie. The server stores its hash.

Recovery

Confirmation and reset codes are single-use. Logout and reset revoke sessions.

Authorization

Financial queries are scoped to the authenticated user. Categories must have the same owner.

Data integrity

The database stores money in integer minor units. Account setup commits as one transaction.

Data minimization

The app does not request bank passwords or card numbers. Payment method is only card or cash.

Technology justification

Choice	Role and reason
Responsive web app	Browser access on iOS, Android and desktop from one codebase.
React + TypeScript + CRA	Reusable screens and typed data. CRA is the requested prototype setup.
Tailwind + Zustand	Consistent responsive styling and a small shared state layer.
Zod + Recharts	Validate form/API data and display spending charts.
Flask + SQLAlchemy + Alembic	A custom API, relational data access and versioned database changes.
SQLite / PostgreSQL option	Simple local persistence with a path to a hosted relational database.
Vercel frontend / Flask API	Frontend demo deployed. Account API needs separate hosting.

Problem and context

Students manage everyday purchases across food, transport, study materials, leisure and other needs. Individual payments may seem small, but remembering them does not provide a reliable view of monthly spending. Purchases made with both cash and cards can be difficult to review together. A bank balance also does not directly explain how much remains within a personal spending plan. TengeFlow addresses this problem through a responsive expense tracker designed for use in Kazakhstan. Users manually record purchases in tenge, assign each expense to one category and compare spending with their chosen monthly budget.

The primary task is to add an expense and immediately understand its effect on the remaining amount. Searchable history, editable entries, category limits and visual insights support continued review. The intended user is a student who is comfortable with everyday mobile apps but has limited time and a modest budget. Clear labels, short forms and recoverable errors support this context. The project demonstrates an ICT solution combining a browser interface, application state, network requests, authenticated access and database storage. It does not connect to banks or claim measured improvements in financial behaviour.

Context developed from the project brief and CFPB guidance [1]. The persona is illustrative, not a research participant.

Evidence and sources

[1] Consumer Financial Protection Bureau

Track your spending with this easy tool. Basis for recording spending and reviewing habits.

[2] BudgetBakers / Wallet Help Center

Setup Budgets. Reference for category limits and avoiding overlapping budgets.

[3] OWASP Cheat Sheet Series

Authentication Cheat Sheet. Reference for authentication and session protection.

[4] Flask documentation

Deploying to Production. Hosted operation requires a production server and deployment setup.

[5] Project evidence

TengeFlow master brief, current source code and actual local UI screenshots.

Sources accessed 2 October 2026. Interface data and personas are fictional. Project illustrations are AI-generated.

Assessment coverage

Required component	Evidence in this PDF
1 Problem and context	2, 17 (150-200 words)
2 Target users and persona	2
3 ICT solution and at least five functions	3-6
4-5 Components and architecture diagram	8, 15-16
6 Network and full data transaction	8-10
7 Cloud or mobile technology	7, 10
8 Main user scenario	4-5, 9
9 Connected prototype and recovery states	3-7, 13-14
10 Technology justification	16
11 Limitations and future development	11

Live demo



Scan to open TengeFlow

<https://ict-midterm-artyom.vercel.app>